

When is a fake concert still a concert?

A study of adjective-noun composition in LLMs

Hayley Ross¹, Kathryn Davidson¹, Najoung Kim²

¹ Harvard University ² Boston University

Correspondence: hayleyross@g.harvard.edu



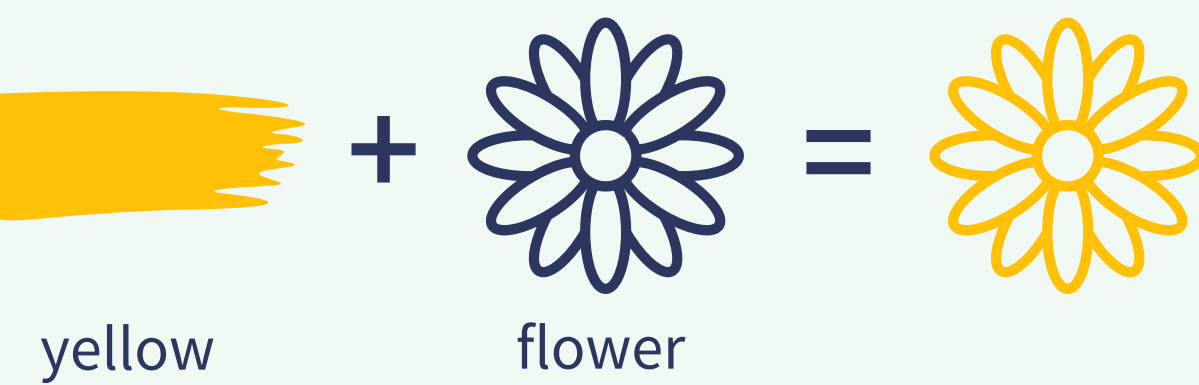
Summary

- LLMs broadly capture privative vs. subjective entailment patterns for adjectives, and can generalize to unseen adjective-noun bigrams
- We propose two ways to evaluate LLMs against a distribution of human data (as opposed to single correct answers)

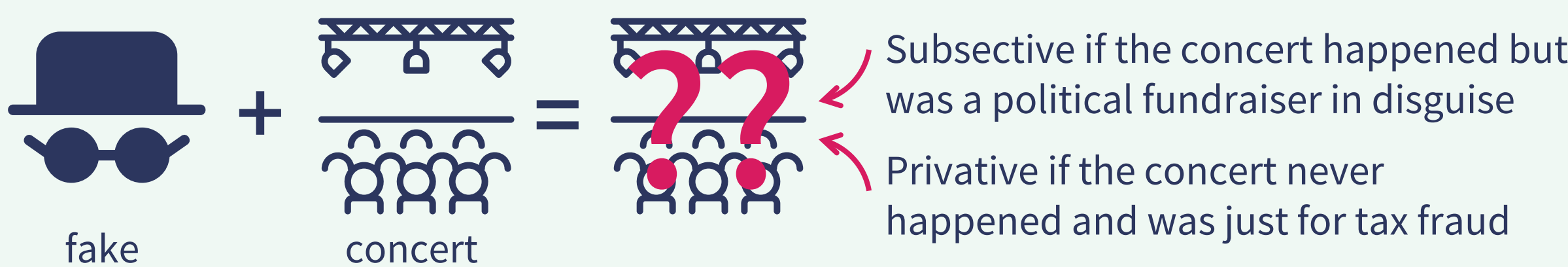
Entailments depend on the adjective, noun and context

Humans (and LLMs) can't memorize the entailment by adjective or adjective-noun pair, they must compute it on the fly:

Subjective entailment

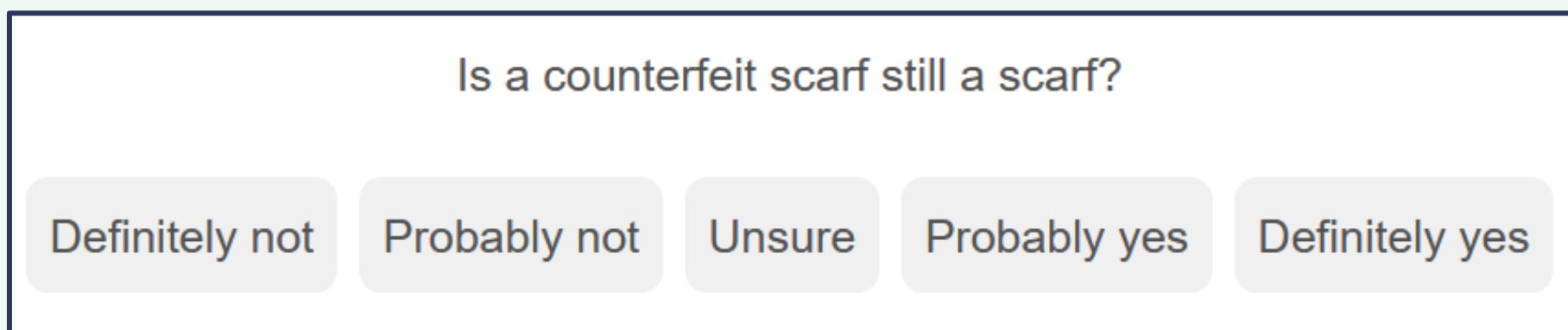


Privative entailment



Experiment design for human judgments

12 ratings per bigram on a 5-point Likert scale; 305 bigrams (57 novel)



Screenshot from human judgment collection experiment

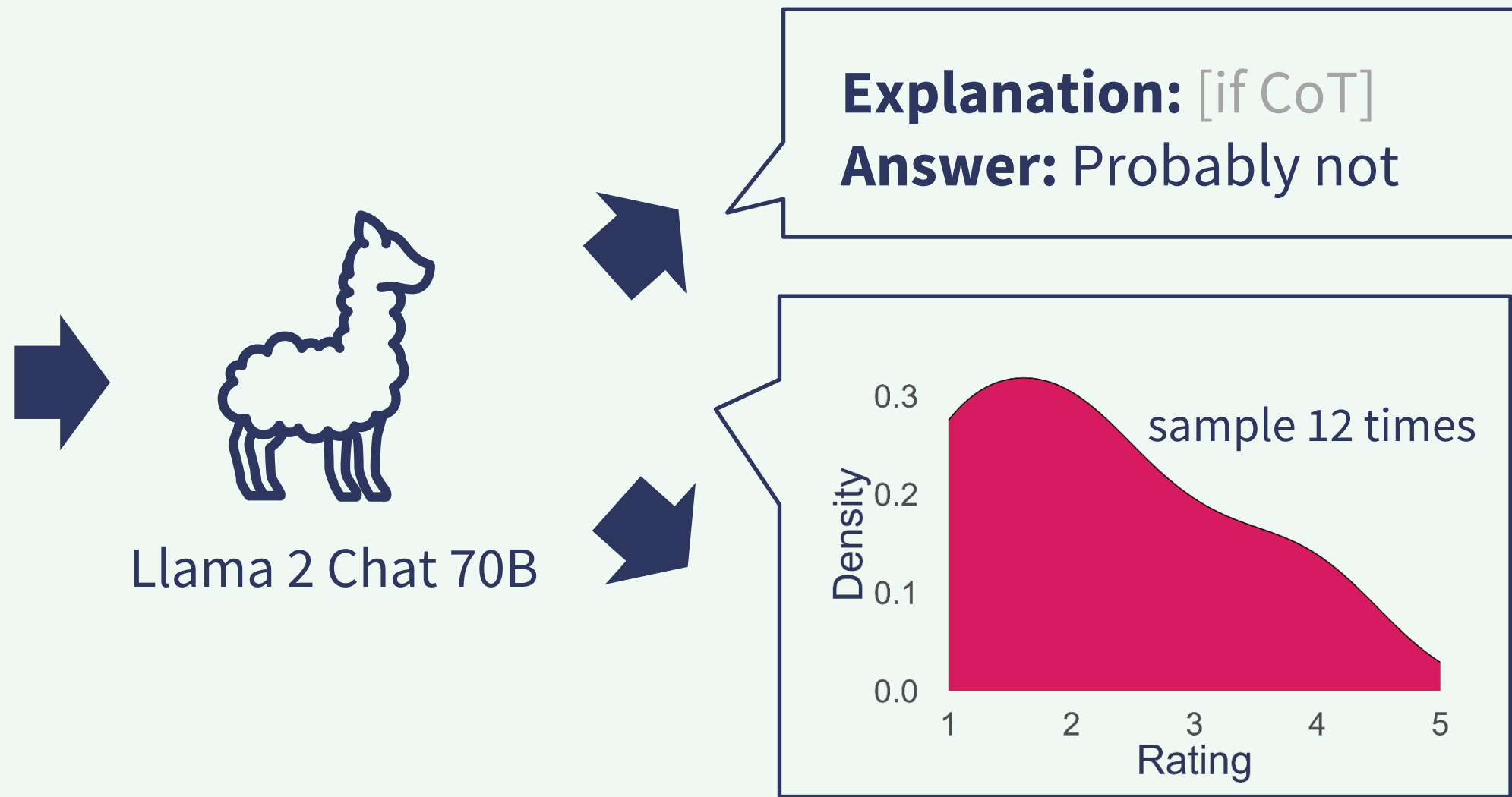
LLM prompt design

[5 few-shot examples]

Question: On a scale of 1 to 5, where 1 is "Definitely not", 2 is "Probably not", 3 is "Unsure", 4 is "Probably yes" and 5 is "Definitely yes", is a fake concert still a concert?

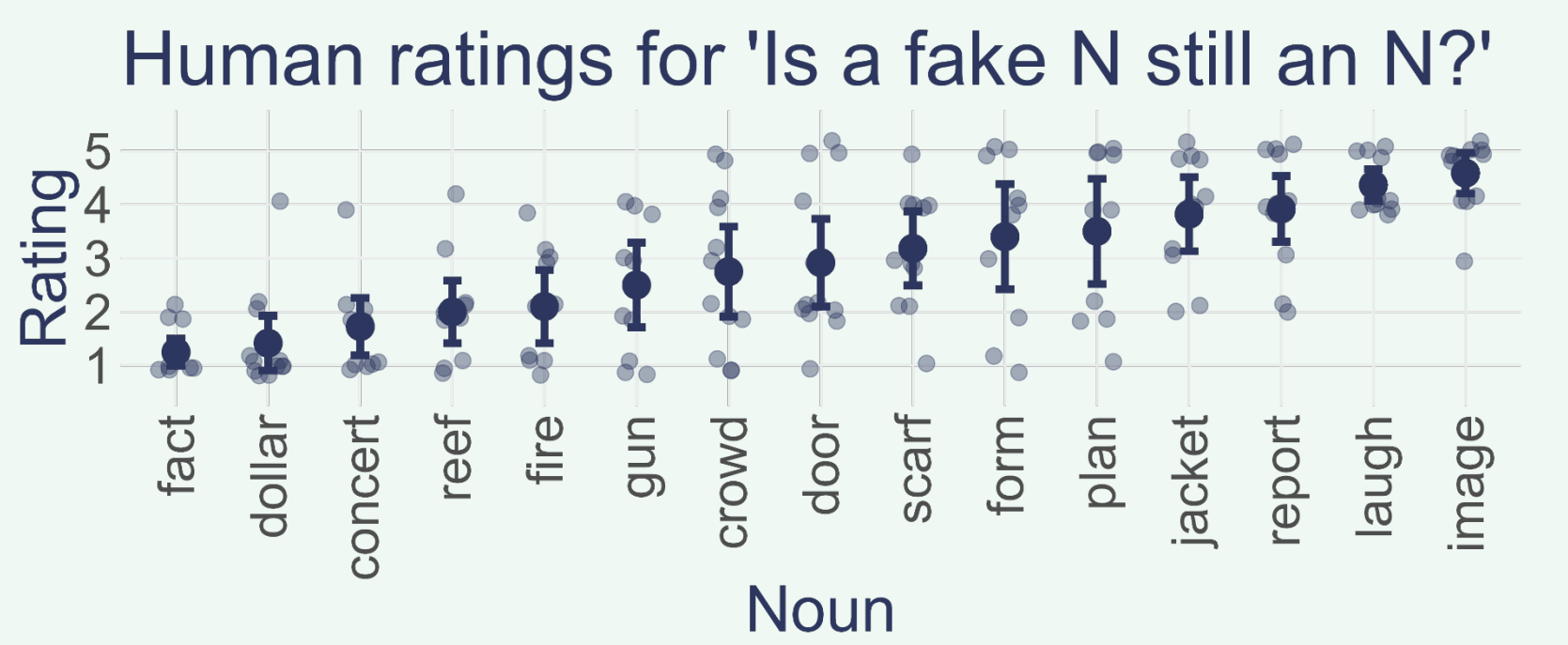
Explanation: [only provided in CoT setting]

Answer: _____



How do we evaluate an LLM against a distribution of ratings?

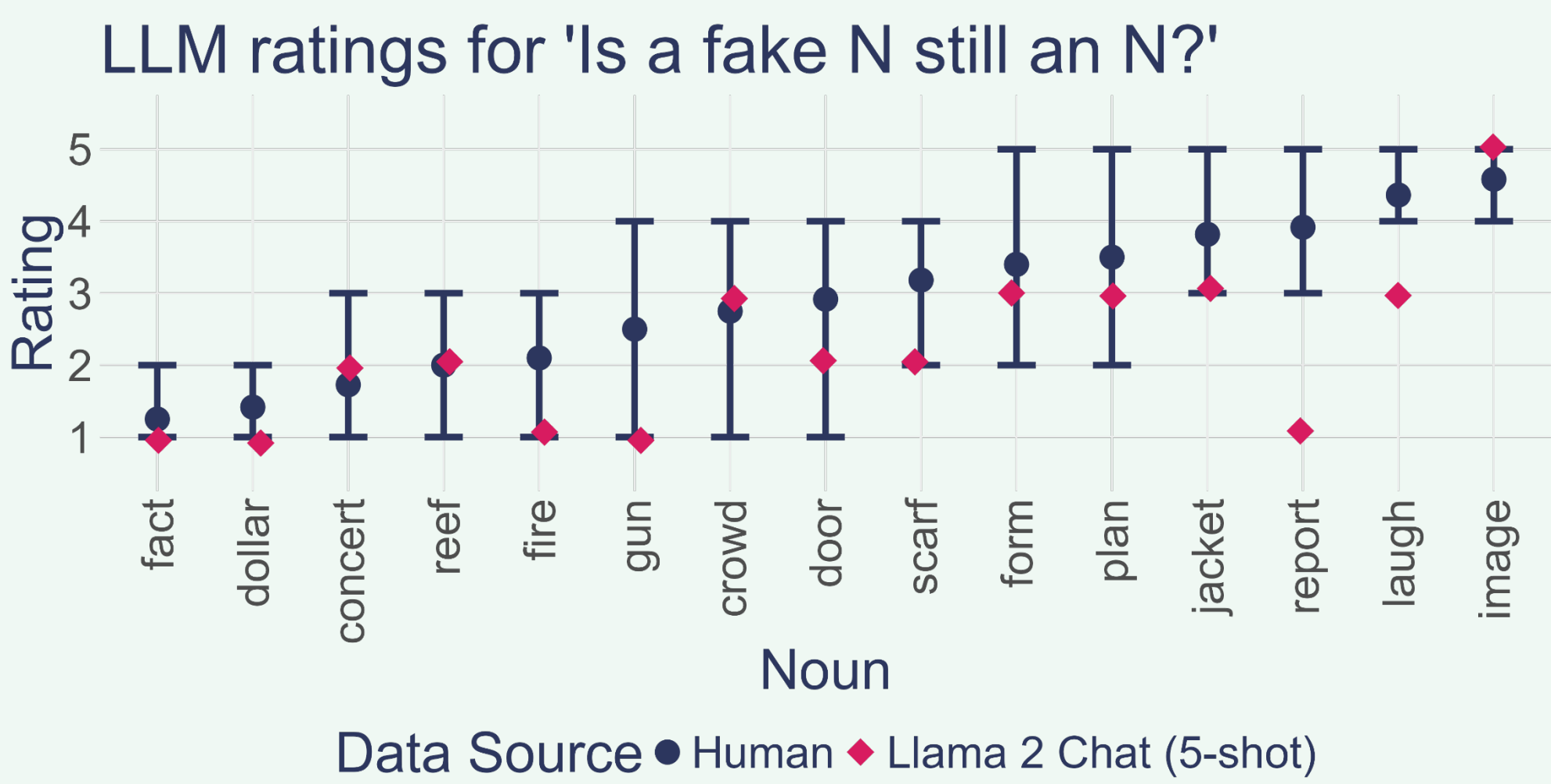
- For many adjective-noun bigrams, the human ratings show high variance; there is no single correct answer
- We can compare a single deterministic prediction from the LLM, or we can sample a distribution from the LLM (either from the log-probs, or using temperature)



Human ratings show high variance for many bigrams

Option 1: Accuracy within 1 SD of human mean

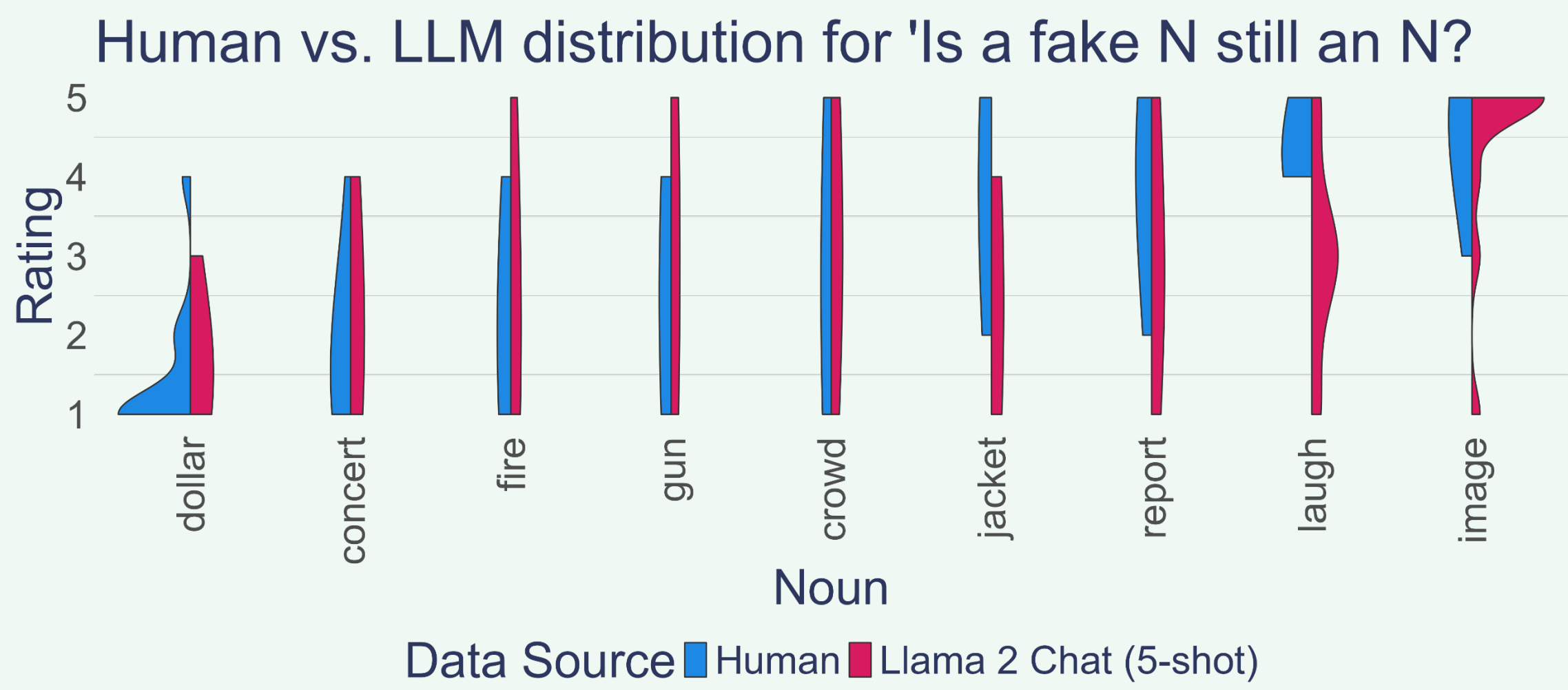
Compares single LLM prediction to human distribution



Llama 2 Chat's ratings for "privative" adjectives are often within 1 SD (rounded) of the human mean, but this is easy for high SDs

Option 2: Distance between distributions

Compares LLM distribution to human distribution



The distribution sampled from Llama 2 Chat's log-probabilities only sometimes agrees with the human rating distribution

Llama 2 Chat broadly captures human entailment patterns and can generalize to novel bigrams

Rating Source	Accuracy within 1 SD of human mean			
	"Privative" Adj.	"Subjective" Adj.	Novel	Overall
Humans (average)	0.90	0.97	0.94	0.93
"Majority" baseline	0.77	0.99	0.91	0.88
Llama 2 Chat, 5-shot CoT	0.72	0.81	0.82	0.76
Llama 2 Chat, 5-shot	0.81	0.90	0.87	0.85
Llama 2 Chat, 0-shot	0.65	0.99	0.77	0.81
Random baseline	0.61	0.32	0.47	0.46

Next steps

- Test LLMs with contexts that bias the entailment and compare to humans
- Sample LLM rating distribution for range of (LLM-generated) contexts
- Theory & analogy-inspired prompting and baseline model(s)
- Evaluate more LLMs & LLM sizes
- Get human ratings for more bigrams

Supported by the Mind Brain Behavior Initiative at Harvard University.
Image credits: Adrien Coquet, Goat Team, Kosong Tuijuh, Pongsakorn, Puspa Kusuma, Ricons, Sam Designs, Selot Lo, Vecto, Zaenal Abidin @ [Noun Project](#)